

Coach Kai API 500 Error - Fixed

Issue Summary

Coach Kai was returning “API error 500 service unavailable” on both mobile and desktop, preventing users from chatting with the AI coach.

Root Cause Analysis

The API endpoint `/app/api/ai-coach/chat/route.ts` was failing due to:

1. **Missing error handling** - Database errors would crash the entire API
2. **No API key validation** - The code didn't check if `ABACUSAI_API_KEY` exists before making requests
3. **Poor error logging** - Generic error messages made it impossible to debug
4. **Network error vulnerability** - External API calls could fail silently

Fixes Applied

1. API Key Validation (Lines 23-30)

```
// Check for API key first
if (!process.env.ABACUSAI_API_KEY) {
  console.error('[Coach Kai] CRITICAL: ABACUSAI_API_KEY is not configured');
  return NextResponse.json(
    { error: "AI service not configured. Please contact support." },
    { status: 503 }
  );
}
```

2. Database Error Protection

Wrapped ALL database operations in try-catch blocks:

-  User conversation creation/fetching (Lines 47-74)
-  User data loading (Lines 80-99)
-  Partner recommendations (Lines 140-163)
-  Message saving (Lines 378-400)

Result: Database failures no longer crash the API - it continues and responds anyway.

3. Enhanced AI API Error Handling (Lines 260-300)

- Network errors caught separately
- Detailed error logging with status, response, and user context
- User-friendly 503 errors instead of generic 500s
- Graceful degradation for regeneration failures

4. Comprehensive Error Logging

All errors now log:

- Error type and message
- Stack trace for debugging

- User ID for troubleshooting
- API response details

Critical: Environment Variable Required

The API requires `ABACUSAI_API_KEY` to be configured in Vercel environment variables.

How to Check/Add on Vercel:

1. Go to <https://vercel.com/silentvector001s-projects/mindful-champion>
2. Click **Settings** → **Environment Variables**
3. Look for `ABACUSAI_API_KEY`
4. If missing, add it:
 - Key: `ABACUSAI_API_KEY`
 - Value: Your Abacus AI API key
 - Environments: Production, Preview, Development
5. **Redeploy** after adding

Where to Get the API Key:

- Abacus AI Dashboard: <https://apps.abacus.ai/>
- API Keys Section: <https://abacus.ai/app/profile/apikey>

Files Modified

-  `/app/api/ai-coach/chat/route.ts` - Enhanced with comprehensive error handling

Deployment Status

-  **Committed:** Commit hash `19cc3f0`
-  **Pushed:** master branch updated
-  **Vercel Build:** Deploying now
-  **Action Required:** Verify `ABACUSAI_API_KEY` is set in Vercel

Testing After Deployment

1. Check Vercel Logs

- Go to Vercel dashboard → Deployments → Latest
- Click “View Function Logs”
- Look for any `[Coach Kai]` error messages

2. Test Coach Kai

1. Visit <https://mindfulchampion.com/train/coach>
2. Click the microphone or type a message
3. Check for responses

3. Expected Behavior

-  **If API key is valid:** Coach Kai responds normally
-  **If API key is missing:** “AI service not configured” error (503)
-  **If API key is invalid:** “Coach Kai is temporarily unavailable” (503)

-  **Database errors:** API still works, just logs warnings

Other APIs Using Same Key

These endpoints also need `ABACUSAI_API_KEY` :

- `/api/ai-coach/recommendations/route.ts`
- `/api/ai-coach/insights/route.ts`
- `/api/ai-coach/route.ts`
- `/api/ai-coach/daily-coaching/route.ts`
- `/api/training/generate-program/route.ts`
- `/api/chat/coach/route.ts`

All will fail if the API key is not configured.

Next Steps

1.  Code fix deployed
2.  **USER ACTION:** Verify `ABACUSAI_API_KEY` in Vercel environment variables
3.  **USER ACTION:** Redeploy if key was just added
4.  Test Coach Kai functionality
5.  Monitor Vercel logs for any remaining errors

Success Metrics

- Coach Kai responds to user messages
- No 500 errors in Vercel logs
- Detailed error messages if something fails
- Graceful degradation (API works even if database has issues)

Commit: `19cc3f0` - "Fix Coach Kai API 500 error with comprehensive error handling"

Date: December 17, 2025

Status:  **DEPLOYED** - Awaiting environment variable verification